Narrated Accessibility Walk

North Cambridge, MA: Massachusetts Ave, Norris St, Shea Rd, Locke St and Woodbridge St

Pilot report 1 | Walk recorded October 3, 2026
Prepared by Uri Kartoun, PhD, Founder, DBbun LLC, Cambridge, MA | uri.kartoun@dbbun.com | dbbun.com

Summary

A six-minute narrated smartphone video of a neighborhood walk was turned into a structured list of possible barriers for people with disabilities. This report uses only the content of the video itself: the footage and the walker's spoken narration.

- **11 possible barriers** were identified: 4 high, 5 moderate and 2 low severity.
- **The most serious problems are on the brick side-street sidewalks:** a utility pole in the walking path, tree-root heaving, and construction cones placed on the sidewalk. Twice the walker noted that a wheelchair user would have to move into the road.
- **Mass Ave was mostly accessible:** wide concrete sidewalks, curb ramps with tactile warning strips and marked crosswalks. The main concern there is fast-turning traffic at Shea Rd and Norris St.
- **No public benches were seen anywhere on the route,** which matters for people with fatigue, pain or other energy-limiting conditions.

Disclaimer. This report was produced with the help of AI tools (automatic speech transcription and AI-assisted image analysis; see "How this report was produced") and was reviewed by the author. Errors and omissions are possible. Items are possible barriers observed in a single video, not legal compliance determinations or findings of violation. Observations reflect conditions on October 3, 2026 and may have changed since.

Walk details

Date and time	October 3, 2026, about 3:49 pm (from video metadata)
Duration	6 minutes 3 seconds; one continuous walk
Route	Mass Ave at Shea Rd / Norris St; Mass Ave sidewalk; Locke St and Woodbridge St; Shea Rd; back to Mass Ave
Recording	iPhone XS, 1280 × 720 video at 30 frames per second, with spoken narration
Narrator	Uri Kartoun, PhD (DBbun LLC); assessed the route from the point of view of wheelchair, walker and cane users and people who tire easily
Conditions	Dry and sunny, early autumn, leaf litter

Possible barriers

Severity: **High** = may stop a wheelchair or walker user, or force them into the street. **Moderate** = slows people or adds effort or tripping risk. **Low** = minor or temporary. Times refer to the video.

#	Time	Location	Possible barrier	Most affected	Severity	Suggested fix
1	00:00–00:24; 05:46–05:57	Mass Ave at Shea Rd / Norris St	Vehicles cross Mass Ave from Norris St and turn quickly into Shea Rd across the crosswalk; little time for slow pedestrians.	Wheelchair and walker users; slow walkers; low vision	High	Leading pedestrian interval; turn calming; signal-timing review
2	02:00–02:06	Brick sidewalk, side street	Missing bricks with a pile of loose bricks and a cone narrowing the walking path.	Wheelchair, walker, low vision	Moderate (temp.)	Complete repair; keep a clear path during work
3	02:30–02:39	Side-street sidewalk (Locke St / Woodbridge St area)	Tree-root heave: lifted, broken slab and an exposed patch of dirt.	Wheelchair, walker, cane	High	Tree-safe repair: flexible or rubber paving, ramp the path over roots, or widen the path or tree pit (avoid harming the tree)
4	03:09–03:15	Corner with curb ramp	Utility pole beside the curb ramp landing; heavily cracked, faded crosswalk paint.	Wheelchair, low vision	Moderate	Re-stripe crosswalk; check landing clear space
5	03:42–03:51	Side street	Temporary no-parking cones and signs placed on the sidewalk next to parked trucks.	Wheelchair, walker, stroller, low vision	Moderate (temp.)	Place cones in the parking lane, not on the sidewalk
6	04:05–04:15	Brick sidewalk at corner near Shea Rd	Utility pole within the walking path at the curb-ramp approach; a wheelchair cannot pass and must use the road.	Wheelchair, walker	High	Relocate the pole or widen the approach
7	03:57–04:45	Brick side-street sidewalks	Narrow passage between fences and signposts or poles; uneven bricks.	Wheelchair, walker, low vision	Moderate	Consolidate signposts; reset bricks
8	04:42	Brick sidewalk near a tree	Heaved bricks (already marked with paint).	Walker, cane, wheelchair	Moderate	Reset bricks with tree-safe methods (flexible base, wider tree pit)
9	05:30–05:39	Shea Rd	Hedge overgrowth narrowing the sidewalk.	Wheelchair, low vision, stroller	Low	Owner trimming; city notice
10	01:06–01:15	Mass Ave curb by parking	Displaced granite curbstone at the curb edge.	People getting in or out of cars; low vision	High (trip risk)	Reset the curbstone
11	00:36–01:03	Mass Ave sidewalk	Cracked, uneven concrete joints with weeds.	Walker, cane, wheelchair	Low	Joint repair

What worked well

- Curb ramps with tactile warning strips at the Mass Ave crossings (00:00–00:24, 01:30–01:39, 05:48).

- Wide, flat concrete sidewalks along Mass Ave.
- Marked crosswalks and a bollard protecting the corner at Mass Ave.

Energy, sensory and environment

- **Rest:** no benches seen on the whole route.
- **Surface effort:** brick sidewalks with heaving require more effort than smooth concrete for wheelchair and walker users.
- **Shade:** partial tree cover on the side streets; Mass Ave is exposed.
- **Traffic stress:** crossing Mass Ave with turning vehicles.
- **Not captured this time:** smells or fragrances, noise levels, night lighting, temperature, and the walker's own effort ratings.

Photo evidence

Frames were taken from the video. Faces and license plates were blurred or are not identifiable.



Figure 1. Mass Ave crossing at Shea Rd / Norris St: curb ramp with tactile warning strip and a bollard (positive), at the location where fast-turning traffic was flagged (barrier 1).



Figure 2. Displaced granite curbstone at the curb edge next to the Mass Ave travel lane (barrier 10).

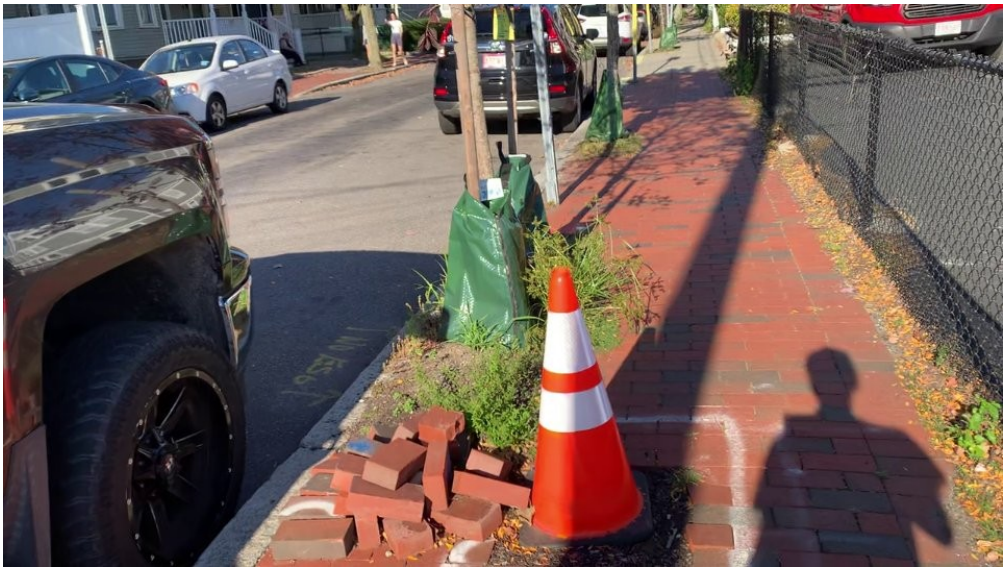


Figure 3. Missing bricks, a pile of loose bricks and a traffic cone narrowing the brick sidewalk (barrier 2).



Figure 4. Tree-root heave with broken pavement and exposed dirt along a picket fence (barrier 3).



Figure 5. Utility pole next to the curb ramp landing and cracked, faded crosswalk markings (barrier 4).



Figure 6. Temporary no-parking cones and signs placed on the sidewalk (barrier 5).



Figure 7. Utility pole within the walking path at the curb-ramp approach; a wheelchair user would need to enter the road (barrier 6).



Figure 8. Heaved bricks near a tree, already marked with paint (barrier 8).

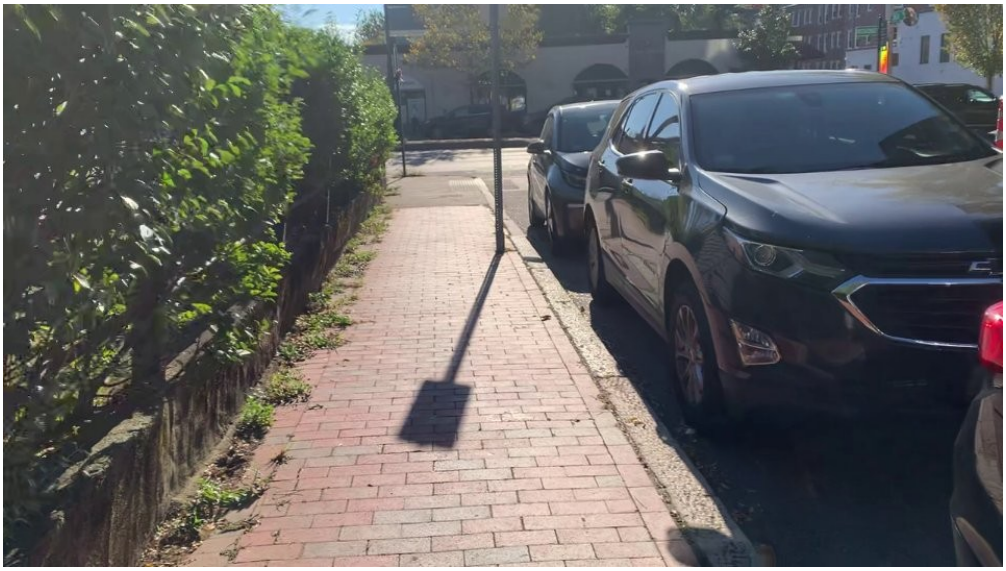


Figure 9. Hedge overgrowth narrowing the sidewalk on Shea Rd (barrier 9).



Figure 10. Returning to Mass Ave: curb ramp with tactile warning strip and marked crosswalk (positive).

How this report was produced

- **Capture and local processing:** one narrated smartphone video, processed on the walker's computer: audio extracted with ffmpeg (imageio-ffmpeg 7.1); narration transcribed with faster-whisper, model medium.en; one frame sampled every 3 seconds (121 frames); faces blurred with OpenCV 4.14 before any image left the computer.
- **AI review:** the blurred frames and the transcript were reviewed by Claude (Anthropic), model claude-opus-5-5, which drafted the barrier list and this report. The walker reviewed and corrected the result, including street names.
- **Human review:** the walker confirmed or rejected each AI-proposed item. 1 of 12 proposed items was rejected: an A-frame sign that stood in the tree pit and curb zone, not in the walking path. Street names misheard in the transcript were also corrected.
- **Community feedback:** after the first version was shared in a neighborhood group, a resident with a disability asked that fixes for tree roots protect the trees, and raised the need for space for larger power wheelchairs and scooters. The suggested fixes for barriers 3 and 8 were revised accordingly.
- **No other data sources** were used in this report.

Limitations

- One walker, one sunny afternoon, one short route; no GPS track, so side-street locations are approximate.
- Automatic face blurring can miss faces; check every image before sharing publicly.
- Items are possible barriers observed in the video, not legal compliance determinations.

Next steps

- Record with location services on (or name each cross street aloud), pause briefly at each issue, and narrate effort ratings and sensory observations such as smell, noise and glare.
- End future walks at a clinic or research site entrance, and invite residents with disabilities to review reports and record their own walks.